Software Test Plan (STP)

Corporate Mental Wellness & Counseling Platform (Problem Statement #14)

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1. Introduction

Purpose: This document is the test plan for the Corporate Mental Wellness and Counseling Platform v1.0. It sets out the objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers registration and identity, the self-assessment and its crisis path, booking, the personal trend view, session notes, aggregate reporting, and the security and privacy rules. The corporate directory, the crisis support service and the notification channel are excluded, apart from the platform's own calls to them.

References: SRS v1.0 and SAD v1.0 for this platform.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), RTM (Requirements Traceability Matrix), pseudo-ID (the random identifier used in place of an employee's name).

2. Test items

- Identity and Pseudonym Service (registration, pseudo-ID, mapping)
- Self-Assessment Service (scoring, bands, crisis path, trend, aggregate report)
- Booking Service (availability, slot listing, reservation, reschedule, session notes)
- Notification Service (confirmations)
- API Gateway (token check, pseudo-ID forwarding)
- Wellness Web Portal

3. Features to be tested

Features mapped to SRS requirement IDs:

- FR-001: Verify the employee once and issue a pseudo-ID
- FR-002: Ten-item self-assessment with score and band
- FR-003: Book only into published slots, reserved in one transaction
- FR-004: Trend of the employee's last six assessments
- FR-005: Crisis helpline and priority slot on a severe score
- FR-015: Aggregate report for the administrator
- NFR-001: Slot list under 2 seconds at peak, correct timezones
- NFR-002: Separate encrypted mapping, no report for groups under ten
- NFR-004: Assessment keeps working when notifications are down

4. Features not to be tested

- The corporate identity provider's own logic (owned by the company IT team)
- The crisis support service beyond the platform's call to it
- Session delivery, therapist licensing and billing (out of scope for v1.0)

5. Test approach / strategy

Levels:

- Unit tests inside each service
- Integration tests across services through the API gateway
- System tests of the full flows, end to end
- Acceptance tests against the SRS acceptance criteria

Types:

- Functional testing of the features in section 3
- Regression testing after each fix
- Performance testing of the slot list under load
- Usability testing of the portal, including screen reader use

Entry criteria: a stable build is deployed, test data is loaded, and the test environment is up.

Exit criteria: 100% of planned test cases executed, 0 critical defects open, and all acceptance criteria met.

5.1 Security validation

- Confirm no screen or log after registration shows a corporate identity
- Confirm the identity mapping is encrypted and reachable only by the Identity service
- Confirm the gateway refuses a request with no valid token
- Confirm an aggregate report for fewer than ten people is blocked
- Confirm all connections use TLS 1.2 or higher

6. Test environment

Hardware: Standard servers or VMs for the four services and the gateway, plus a computer and a phone with a current browser for the portal.

Software: Platform build v1.0 (portal, API gateway, four services and their stores), with stubs for the corporate directory, crisis support service and notification channel.

Tools: Postman (API), JMeter (performance), a screen reader (accessibility), Jira (defect tracking).

Test data: Dummy employees, pseudo-IDs, published slots, and sample assessment answers covering every band.

7. Test schedule

Milestones:

- Test case design: 01-Nov-2026
- Environment setup: 03-Nov-2026
- Test execution start: 04-Nov-2026

- Test execution end: 14-Nov-2026
- UAT: 15-Nov-2026 to 17-Nov-2026

8. Test deliverables

- Test plan (this document)
- Test cases (manual and automated)
- Test scripts
- Test data
- Test execution logs
- Defect reports
- Test summary report

9. Roles and responsibilities

Role	Name	Responsibility
QA Lead	Dhanush Shankar Reddy (PES1UG24AM083)	Prepare the plan, coordinate execution
Test Engineer	Aamer Khan (PES1UG24AM902), Bhuvan Kumar P (PES1UG24AM069)	Design and execute test cases, log defects
Developer	Vivek Varma (PES1UG24AM070)	Support defect fixes and triage
Product Owner	Course Coordinator	Approve test results, sign off readiness

10. Risks and mitigation

Risk	Mitigation
Delay in stable build delivery	Ask for early smoke builds and test the stable services first
Test environment downtime	Keep a backup environment on a cloud VM
Stubs for external services behave unlike the real ones	Keep the stubs close to the documented interfaces and note any gaps

11. Assumptions & dependencies

- The stubs for the corporate directory, crisis support service and notification channel are available and stable
- Test data is provided before execution starts
- A stable build of the four services and the gateway is deployed to the test environment

12. Suspension & resumption criteria

Suspend testing if:

- The environment is unavailable for more than 4 hours

- The build is too unstable and blocks more than 30% of test cases

Resume testing if:

- Blocking defects are resolved
- The environment is stable again

13. Test case management & traceability

The RTM in the SRS maps each requirement to its test cases. For example:

- FR-001 (verify once, pseudo-ID) → TC-REG-01
- FR-003 (book published slot) → TC-BK-01, TC-BK-02
- FR-005 (crisis on severe score) → TC-ASM-04
- NFR-001 (slot list under 2 s) → TC-PERF-01, TC-TZ-01
- NFR-002 (separate encrypted mapping) → TC-SEC-01, TC-RPT-02

14. Test metrics & reporting

Metrics collected:

- % of test cases executed
- % passed and failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final test summary report

15. Approvals

Role	Name	Signature / Date
QA Lead	Dhanush Shankar Reddy (PES1UG24AM083)	
Dev Lead	Vivek Varma (PES1UG24AM070)	
Product Owner	Course Coordinator	